



HALO MEGABOARD[GEN2] DEV

DESIGN INPUT 1 TO BA

05.02.25

ATHER

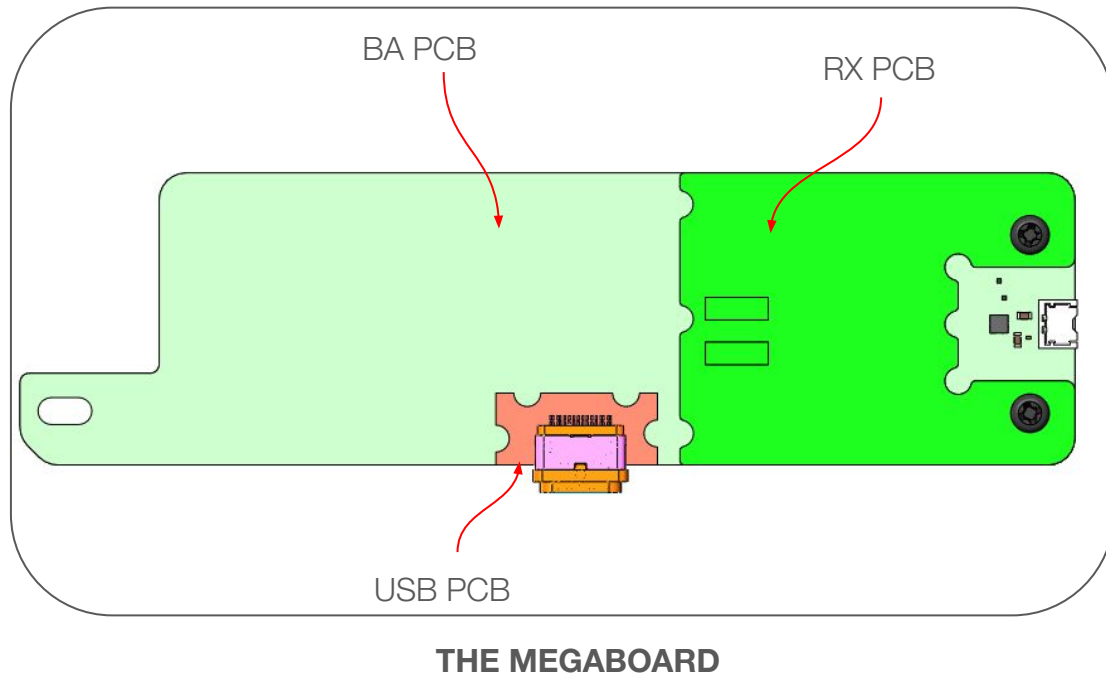
ARCHITECTURE REVAMP

STACKED PCB DESIGN

In the Gen 2 halo, we are willing to switch to a stackable pcb design where all the PCBs are mounted on a single part instead of split strategy.

Why?

Current assembly is time consuming and leads to multiple assembly processes and interconnections of Main board with RX happens using a pigtail. This limits the assembly and is a big bottleneck in streamlining the automated gluing process. Additionally, we want to get rid of the potting process.



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GEN 2 STACKED PCB CONCEPT

Concept

Castellated Holes Design

The castellated holes/ vias are created using a normal via process, which includes Drilling holes and copper plating. These holes look like semi-circles on the edges of the board since they are cut through to form a partial hole or half hole to create an opening on the board edges. Castellated holes can be created in various ways:



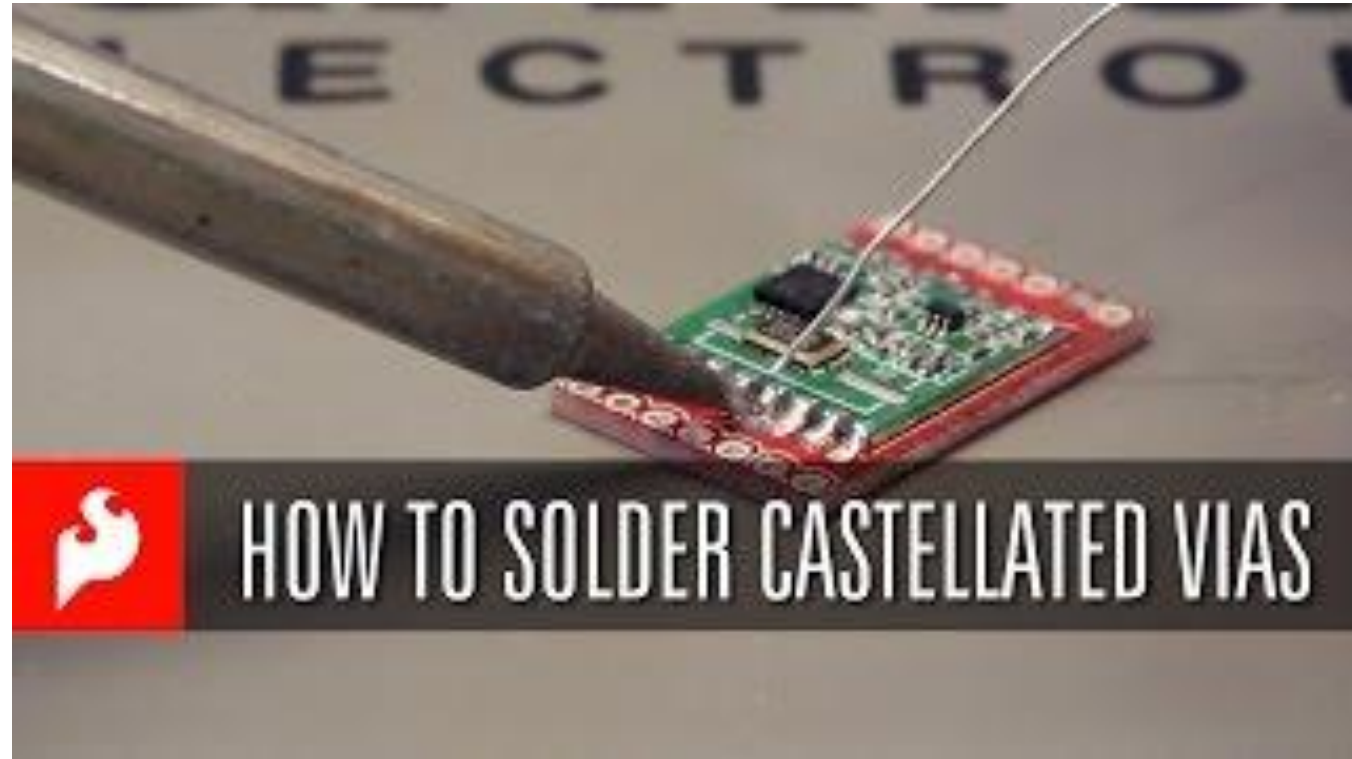
Holes with Smaller cut-in Half Holes



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GEN 2 STACKED PCB CONCEPT

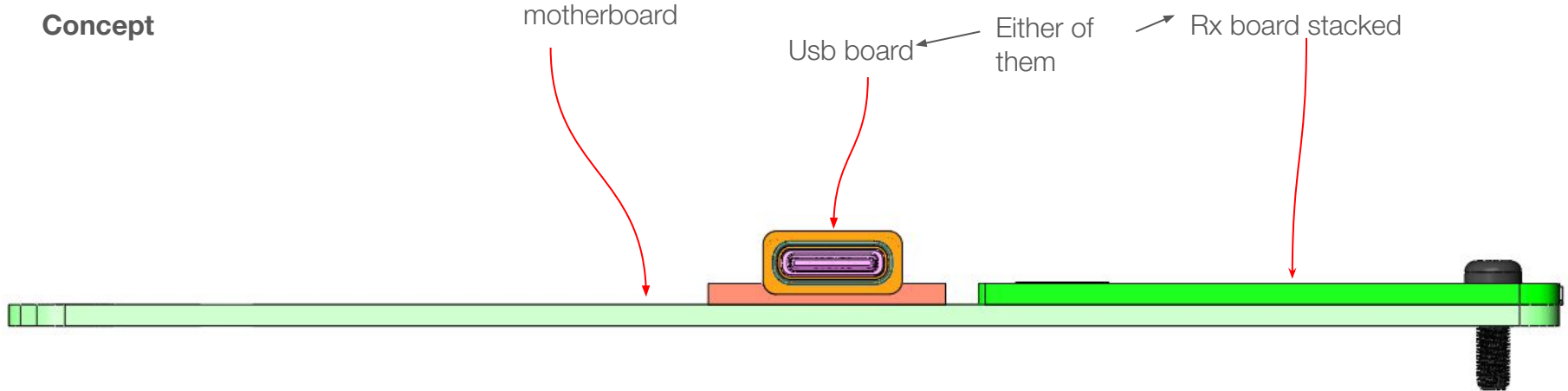
Concept



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THE MEGABOARD CONCEPT

Concept

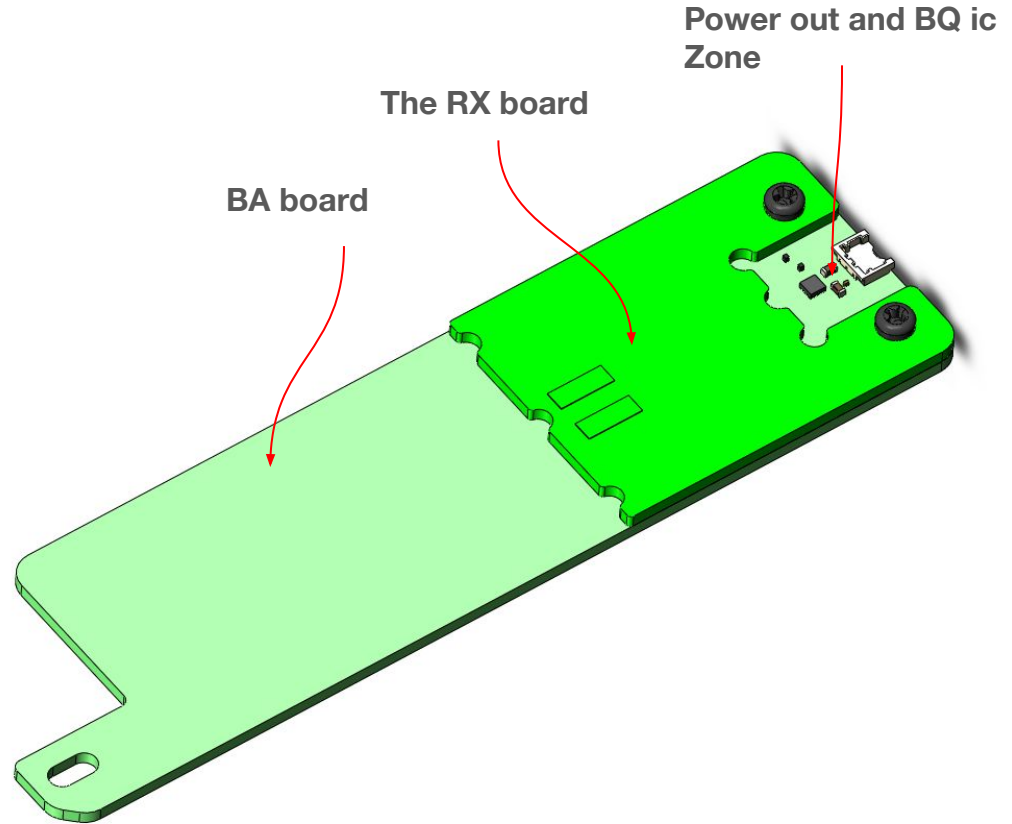


The main board will accommodate either of the boards based on the variants.
USB variant won't have the rx board stacked and the wireless version won't have the USB board populated.

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THE MEGABOARD CONCEPT

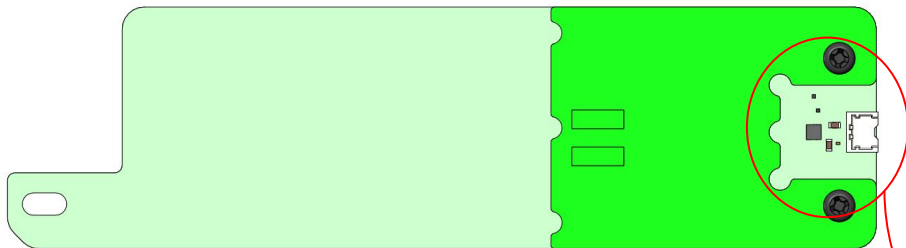
The wireless version



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THE MEGABOARD CONCEPT

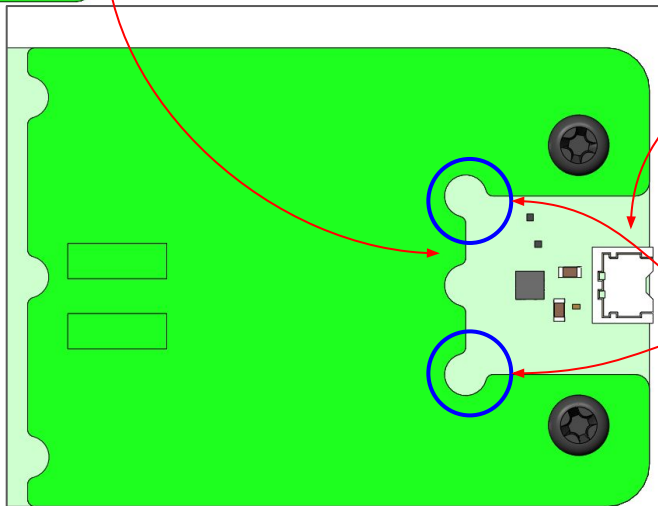
The wireless version



Power out and BQ ic Zone.

The cutout on the RX board is provided to accommodate for the battery connectors and the BQ IC circuitry for minimal power losses and efficiency.

RX board would also have the power termination in the same area.

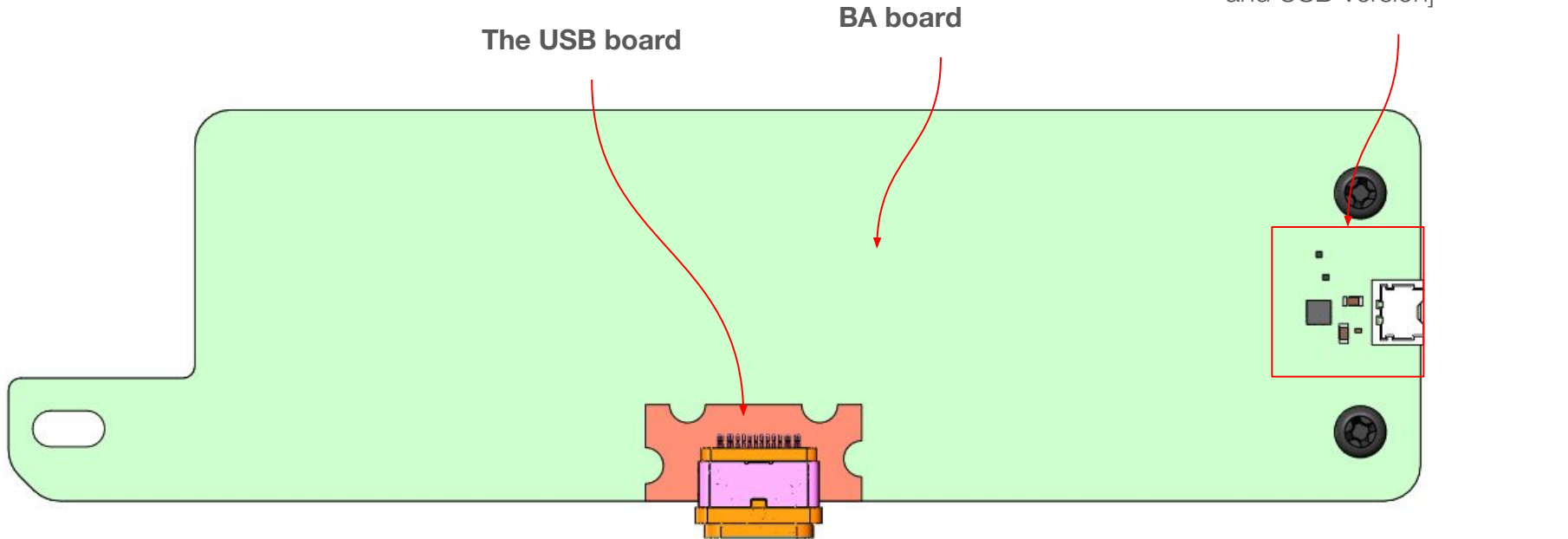


Power out from RX Board
Castellated hole concept

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THE MEGABOARD CONCEPT

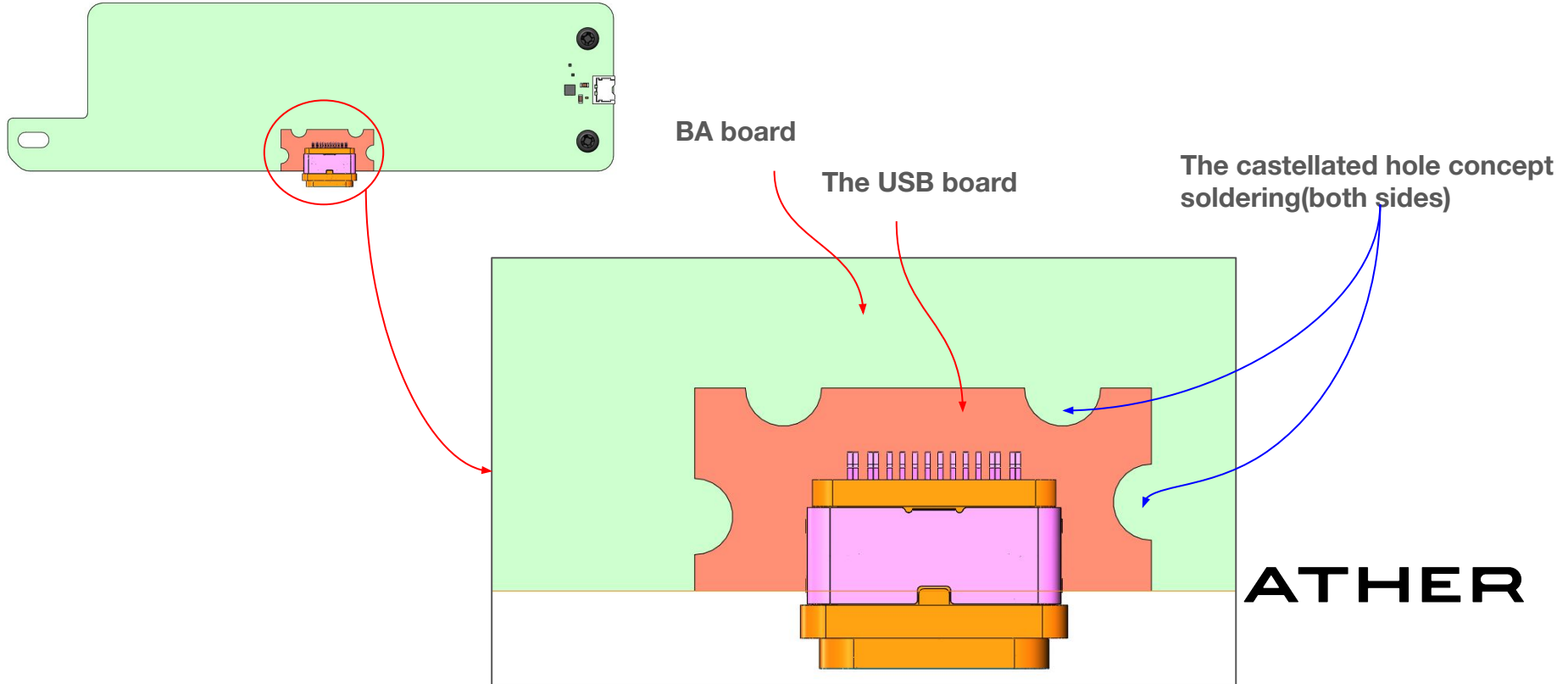
The USB version



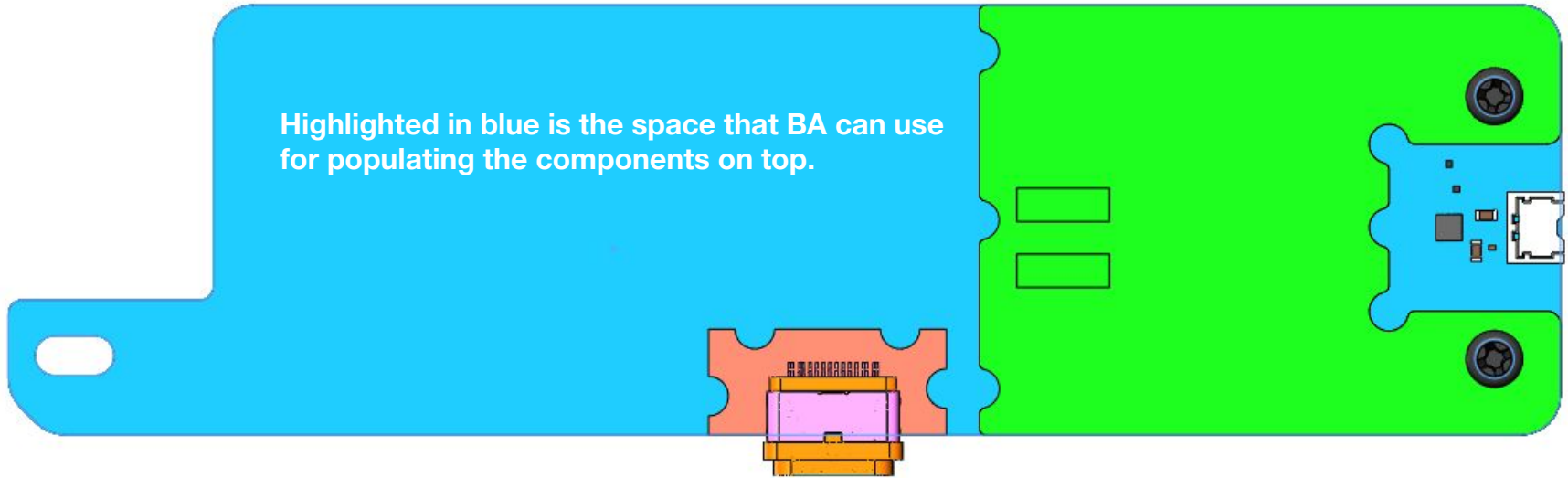
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THE MEGABOARD CONCEPT

The USB version



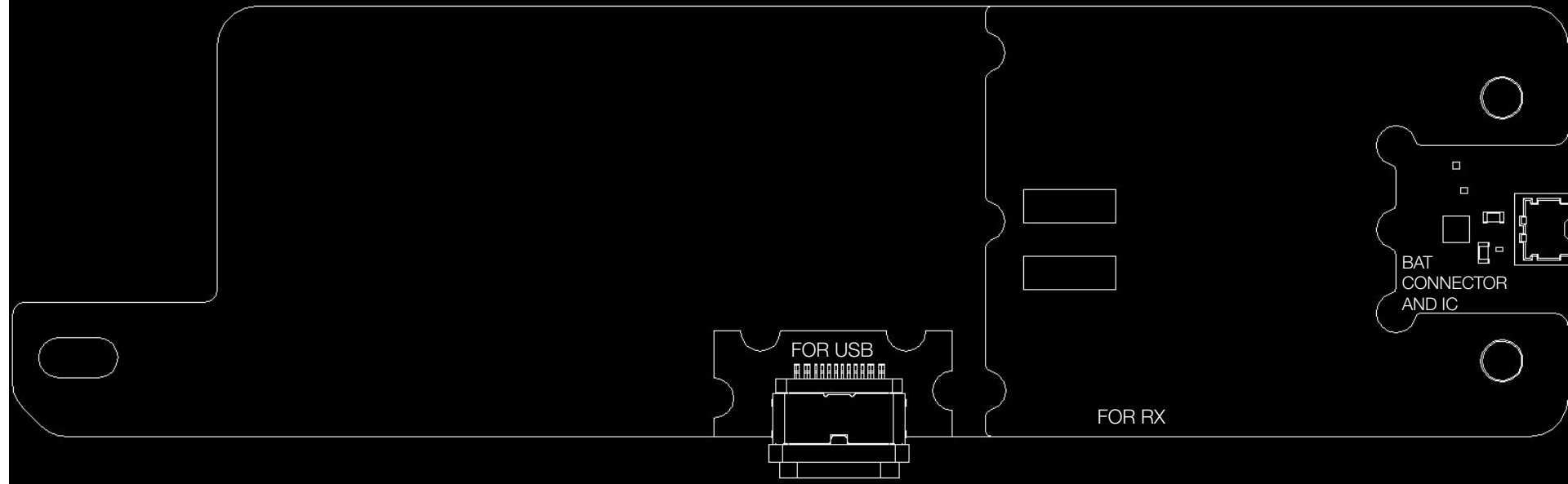
THE MEGABOARD CONCEPT



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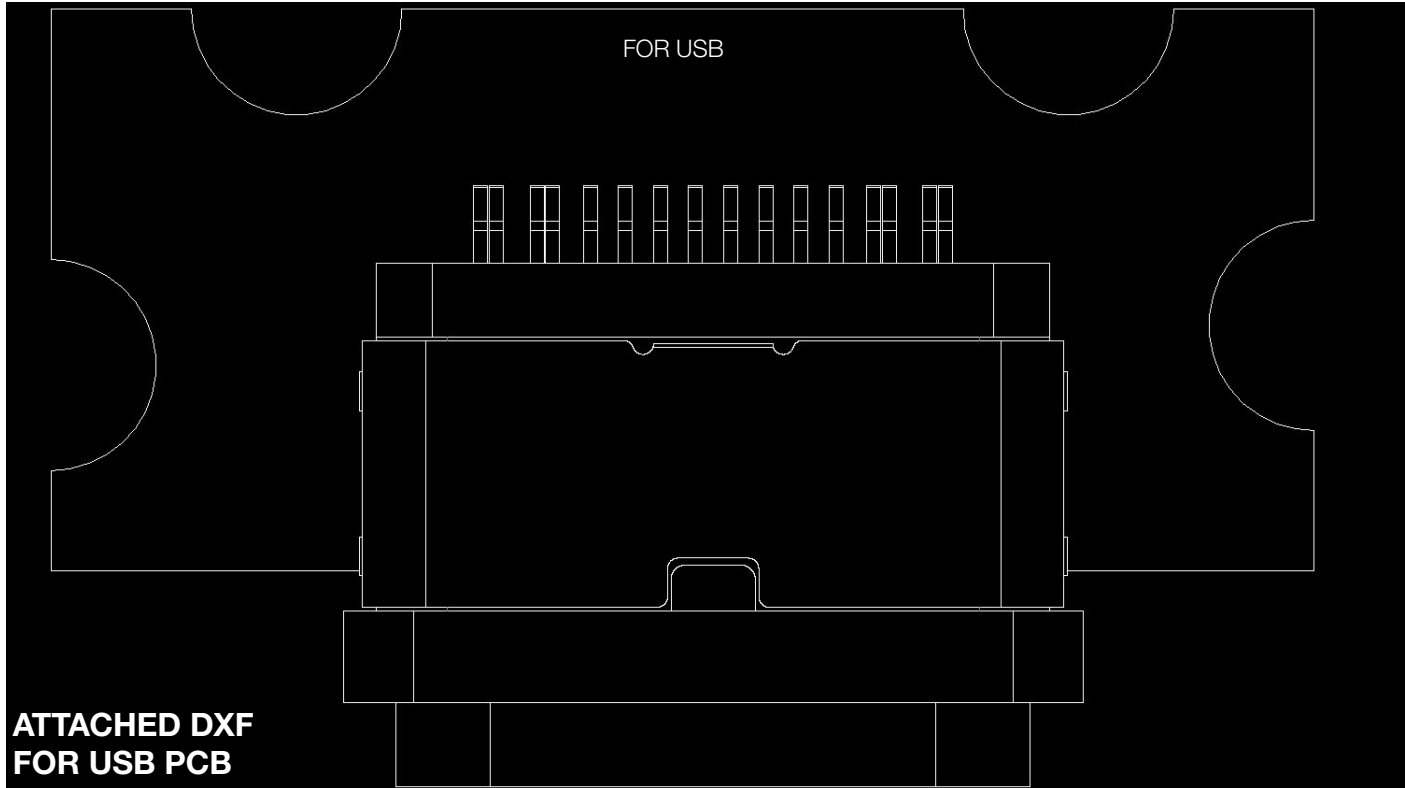
THE MEGABOARD DXF

ATTACHED DXF



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THE USB PCB DXF



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